

Professional Profile

Josep Sanz Campderrós

June 2025

Contents

1	Personal Data	4
2	Academic training	4
3	Professional experience	4
3.1	AIRBUS DS GEO SG, S.A. (Since 2025)	4
3.2	European Home Deliveries (2024-2025)	4
3.3	Freelance (2022-2025)	4
3.4	AB Custom Transports & Logistics, S.L. (2019-2022)	4
3.5	Institut d'Estudis Espacials de Catalunya (2011-2019)	5
3.6	Wide Spectrum Software Solutions (2007-2011 in R+D Dept.)	5
3.7	Enfasystem (2006-2007 in R+D Dept.)	5
3.8	Ingeniería de la Información, IN2 (2005-2006 as team's chief)	5
3.9	Institut d'Estudis Espacials de Catalunya (2001-2005)	5
3.10	Univ. Politècnica de Catalunya, Dept. of Applied Physics (1999-2001)	6
4	About me	6
5	Languages	6
6	Keywords	6
7	Annex (detailed professional experience)	8
7.1	AIRBUS DS GEO SG, S.A. (Since 2025)	8
7.2	European Home Deliveries (2024-2025)	8
7.3	Freelance (2022-2025)	8
7.4	AB Custom Transports & Logistics, S.L. (2019-2022)	9
7.4.1	Last mile mobile application project	9
7.4.2	Online appointment project	10
7.4.3	Label printers project using Raspberry Pi	11
7.4.4	SQLServer replica project in MariaDB	12
7.4.5	Project upload documentation	12
7.4.6	Client integrations project	12
7.5	Institut d'Estudis Espacials de Catalunya (2011-2019)	12
7.5.1	IEEC website project	12
7.5.2	ICE website project	13
7.5.3	IEEC-CSIC telescope project	13
7.5.4	Allsky camera project	14

7.5.5	SQT telescope project	14
7.5.6	IndiCCD driver project	15
7.5.7	MUR project	15
7.5.8	Proyecto OAdM/TJO	15
7.5.9	OpenROCS v2.0 project	16
7.6	Wide Spectrum Software Solutions (2007-2011 in R+D Dept.)	17
7.6.1	NTE/ESA - LTPS project	17
7.6.2	IEEC - GOLD_RTR v2.0 project	17
7.6.3	IEEC - GOLD_RTR live CD project	17
7.6.4	IEEC - GOLD_RTR v3.0 project	18
7.6.5	ISEC Auditors - MANAGER WIPS project	18
7.6.6	AXA Winterthur - Tracking project	18
7.6.7	VUELING - PUNTO 2 project	18
7.6.8	AIDA CENTRE - PDA HORMIPRESA project	19
7.6.9	WS3 - SaltOS Project (www.saltos.org)	19
7.6.10	WS3 - RhinOS Project (www.saltos.org/rhinos)	20
7.6.11	CVA - SISAC project	21
7.6.12	IEEC - GOLD-PRO project	22
7.6.13	RETEVISION - TRACEBOX Audit	22
7.6.14	Alliaria (IN2 GROUP) - FIR@KEY project	22
7.6.15	Shop online CADAICO	22
7.6.16	Web site for the EUSS school	22
7.6.17	Web site for the COACB company	22
7.6.18	Web site for the MEDIATORS OF INSURANCE ASSOCIATION	22
7.6.19	Web site for the VERTEX company	22
7.6.20	Web site for the APQ company	23
7.6.21	Web site and online shop for the SEVIBE company	23
7.6.22	Web site for the TRITON restaurant	23
7.6.23	Web site for the Castelldetels's hostelry school	23
7.6.24	Online shop for the TUTIENDEVIDEOJUEGOS.COM company	24
7.7	Enfasystem (2006-2007 in R+D Dept.)	24
7.7.1	ADMIN project (www.saltos.org/rhinos)	24
7.7.2	Mecano WEB project (www.saltos.org/rhinos)	25
7.7.3	DBMailer project (www.saltos.org/rhinos)	25
7.7.4	EXIT Foundation - Social participation project	25
7.7.5	IEEC - GOLD_RTR project	25
7.7.6	IEEC - PARIS signal processor subsystem project	25

7.8	Ingeniería de la Información, IN2 (2005-2006 as team's chief)	26
7.8.1	SNIFFER project	26
7.8.2	IVP project	26
7.8.3	WEB CONTROL project	26
7.8.4	SICAT project	26
7.8.5	Upgrades for some applications at BAMESA	27
7.9	Institut d'Estudis Espacials de Catalunya (2001-2005)	27
7.9.1	REALTIME (COST716) project	27
7.9.2	DD_RCVR and LOTTOS projects	28
7.9.3	GRACE-II project	28
7.9.4	STD/IEEC Group Intranet	28
7.9.5	GOLD_RTR (PARIS) and MDPP3 (SMOS) projects	29
7.10	Univ. Politècnica de Catalunya, Dept. of Applied Physics (1999-2001)	30
7.10.1	SIGMA, Sistema Informàtic de Gestio i Modelització Acústica	30
7.10.2	NivAval, Nivell d'Avaluació Lar	31

1. Personal Data

- Josep Sanz Campderrós.
- Born in Barcelona on December 7, 1976.
- Email: josep.sanz@saltos.org
- Web: www.josepsanz.net & www.saltos.org
- Github: github.com/josepsanzcamp
- LinkedIn: linkedin.com/in/josepsanzcamp



2. Academic training

- Computer Engineer (UPC).
- Degree of Computer Applications Development.
- Technician in Telecommunications Electronics.

3. Professional experience

3.1. AIRBUS DS GEO SG, S.A. (Since 2025)

Full Stack Software Engineer

3.2. European Home Deliveries (2024-2025)

Developments for GNU/Linux environments and integration of SaltOS 4 in all business areas:

- Integration of SaltOS 4 in all business areas of the company.

3.3. Freelance (2022-2025)

Developments for GNU/Linux and evolutes in the SaltOS and RhinOS projects:

- RhinOS and SaltOS evolutions to run on the latest releases of PHP + third-party library updates.
- Development of the new SaltOS 4 framework: backend in PHP, frontend in JavaScript, API based on REST and JSON requests.
- Migration of SaltOS Server from CentOS 7 to the latest releases of Debian.

3.4. AB Custom Transports & Logistics, S.L. (2019-2022)

Developments for GNU/Linux environments and integration of SaltOS in all business areas:

- Last mile mobile application for drivers, to operate even without internet.
- Route optimization using AI and online appointment automation.
- Integration of SaltOS in all the business areas of the company.
- SaltOS connection with the company's old ERP (SQLServer).
- Documentation recognition systems using OCR, bar codes and QR codes.

- Automation of processes using embedded systems based on Raspberry Pi.

3.5. Institut d'Estudis Espacials de Catalunya (2011-2019)

Developments for GNU/Linux environments to control robotic telescopes:

- Development of OpenROCS 2.0 (GPL-3.0 license), control software used by the TJO and SQT telescopes.
- Development of the www.oadm.cat for the divulgation of the TJO telescope, the www.ice.csic.es for the ICE and the www.ieec.cat for the IEEC.
- Development of the MUR application: an online software for sending astronomical proposals.
- Fork of the IndiCCD project to allow the control of multiple Andor cameras.
- User interface and control system for the SQT telescope, the Allsky camera and the IEEC-CSIC telescope.

3.6. Wide Spectrum Software Solutions (2007-2011 in R+D Dept.)

Developments for GNU/Linux, MONO, Firmware, PDAs and web environments:

- Development of the SaltOS and RhinOS projects released under the GPL-3.0 license.
- Project for the NTE/ESA to do some parts of the unit-test of the LISA Pathfinder project.
- Project for the IEEC to make improvements in the GOLD-RTR project (v2 and v3).
- Multiple web projects (portals and online shops) to different customers.

3.7. Enfasystem (2006-2007 in R+D Dept.)

Developments for WEB using LAMP environments (Linux+Apache+MySQL+PHP):

- Improvement of the 'admin' control panel
- Development of the 'mecano' project
- Development of the DBMailer project
- Several collaborations with IEEC for the PARIS and GOLD-RTR projects

3.8. Ingeniería de la Información, IN2 (2005-2006 as team's chief)

Software development for GNU/Linux and Windows CE.

- Use of security tools like Snort.
- Programming of mobile devices using EVC++ 3.0 for PDA devices
- Programming in C, PHP, .NET
- Use of DBMS such as MySQL, ORACLE

3.9. Institut d'Estudis Espacials de Catalunya (2001-2005)

Development of software for UNIX environments.

- Design and implementation of hardware and drivers.
- Planning and implementation of software for various projects.
- Programming in C with GTK, Tcl/Tk on Unix tcsh.

Administration of Unix Systems and Microsoft Windows

- Implementation of network security systems (SSL)
- Security audits for the computer systems of the department
- Software development and maintenance of the department's intranet
- Software development for the Generalitat de Catalunya (SIGMA and NivAval)
- Collaborations with the WFAAE association.
- Support to users, and so on.

4. About me

- Throughout my professional career I have carried out R+D+i tasks (Research, Development and Innovation).
- I have been working in Unix and GNU/Linux environments for 25 years, I have participated in process automation and big data processing projects (OpenROCS, motor control and astronomical telescopes), backend and frontend development (SaltOS, RhinOS and custom apps for web and desktop) and programming for embedded systems (Altera, Xilinx, Arduino, Raspberry Pi and Trio Motion).
- In the Wide Spectrum Software Solutions business project, I directed the [SaltOS](#) and [RhinOS](#) projects, as well as their release under the GPL-3.0 license, currently I continue developing on these projects.
- I have been a member of the board of directors of CatPL, the association of Catalan Free Software Companies, and I have given talks on free software.
- I have worked in research environments such as the Institut d'Estudis Espacials de Catalunya and the Univ. Politècnica de Catalunya, Dept. of Applied Physics.
- In the last years I have carried out tasks of optimization and automation of processes in the logistics sector, using algorithms such as STRIPS and languages such as PDDL.
- I have published a repository at [github](#) with a web of scores and audios for gralla at [gralla.josepsanz.net](#)
- I have also contributed to free software projects such as phpMyAdmin, Mroonga, MuseScore, PHP, jQuery-UI, jquery-timepicker, TCPDF, WarnockPDF, root-ro, handsontable, masonry, hhvm, jGrowl, sqlite-for-piwik, webapp-xul-wrapper, php-osx, jsmin-php, jquery-grab-bag, pdf.js, Columnizer-jQuery-Plugin

5. Languages

- Spanish native (read, written and spoken).
- Catalan native (read, written and spoken).
- Technical English (intermediate level for read, write and speak).

6. Keywords

- Operating systems: GNU/Linux, Unix Solaris, HP-UX, Slackware, Centos, Debian, Ubuntu, RedHat, Fedora, LAMP, Microsoft, Raspbian
- Databases: MySQL, PostgreSQL, Oracle, SQLServer, MariaDB
- Services: Samba, Apache, FTP, HTTP, SSH, Crontab
- Programming: Visual Basic, Visual C++, C, GTK, PHP, TCSH, TCL/TK, Latex, Embedded Visual C++, Shell Scripts, Python, C++, .NET, XSLT, ANSI C, MONO, WEB, HTML, JavaScript, CSS, Ajax, Vanilla JS

- Programs: Wordpress, EzPublish, VLC, Snort, GIS, Adobe OnSiteView, TomTom, JQuery-UI, Bootstrap, TinyMCE, CKEditor
- Embedded: PLC, FPGA, Altera, TETRA, PDA, NIOS, Xilinx, MicroBlaze, LEON3, Firmware, Raspberry Pi, RPI, RTOS, Trio Motion
- Projects: SaltOS, RhinOS, OpenROCS
- Licenses: GPL, MIT, BSD, Open Source, Free Software
- Logistics: CitasWEB, Maps, Rutas, Transfers, Tracking, Última milla, Entregas, Logística inversa, Recogidas, B2B, B2C, C2C, Orders, Parcels, IA, Call Center, ERP, CMS, OCR, QR, Codebar
- Protocols: TCP/IP, UDP/IP, SSL, Webservice, REST, SNMP, NMEA, RFID, DMA, SOAP, DHCP, API, SMS, Email
- Integration: CSV, Excel, XML, Bytes, EDI, PDF, JPEG, TIFF, XML, RSS, ATOM, JSON
- Technologies: Bluetooth, GPS, Ethernet, GRPS, WIFI, RS232
- Quality: Unit test, Validation test, Cantata, Bazaar, Subversion, Git, phpcs, phpunit, phplloc, phpmmd, phpdod
- Astronomy: Vaisala, INDI, Rain detector, Previstorm, Cloud Sensor, Davis Weather Station, PbCdIComm, XmCCD, IndiCCD, Andor, Finger Lakes Instruments, APOGEE, APC, ROS, SBIG, Meade Telescope, Baader Dome, Allsky
- Others: Reports, Real Time, Oscilloscope, LOPD, Frontend, Backend, Parser, MVC, Mailing, Online, E-Learning, SDK, Auditoria, DoS

7. Annex (detailed professional experience)

7.1. AIRBUS DS GEO SG, S.A. (Since 2025)

Full Stack Software Engineer

7.2. European Home Deliveries (2024-2025)

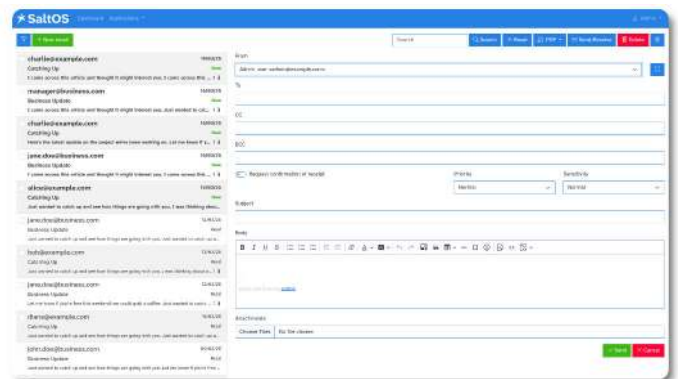
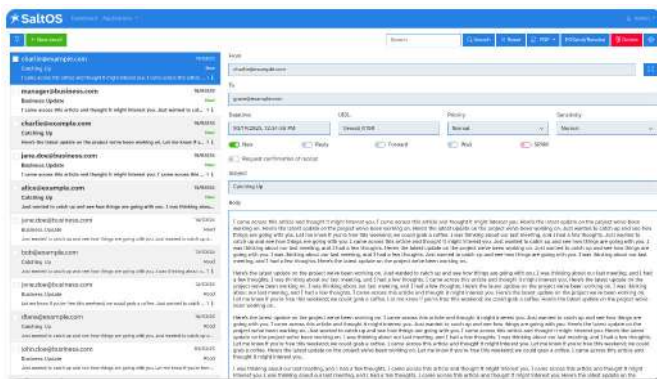
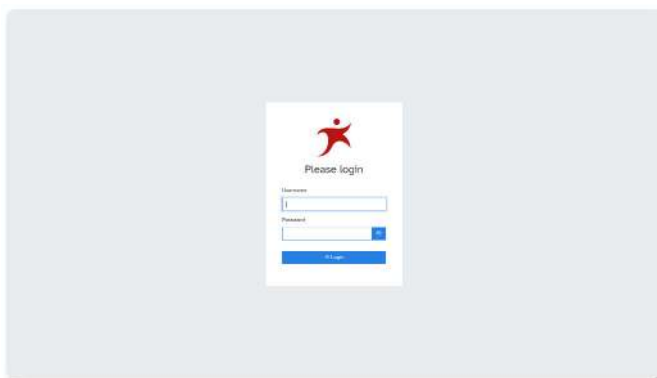
Developments for GNU/Linux environments and integration of SaltOS 4 in all business areas:

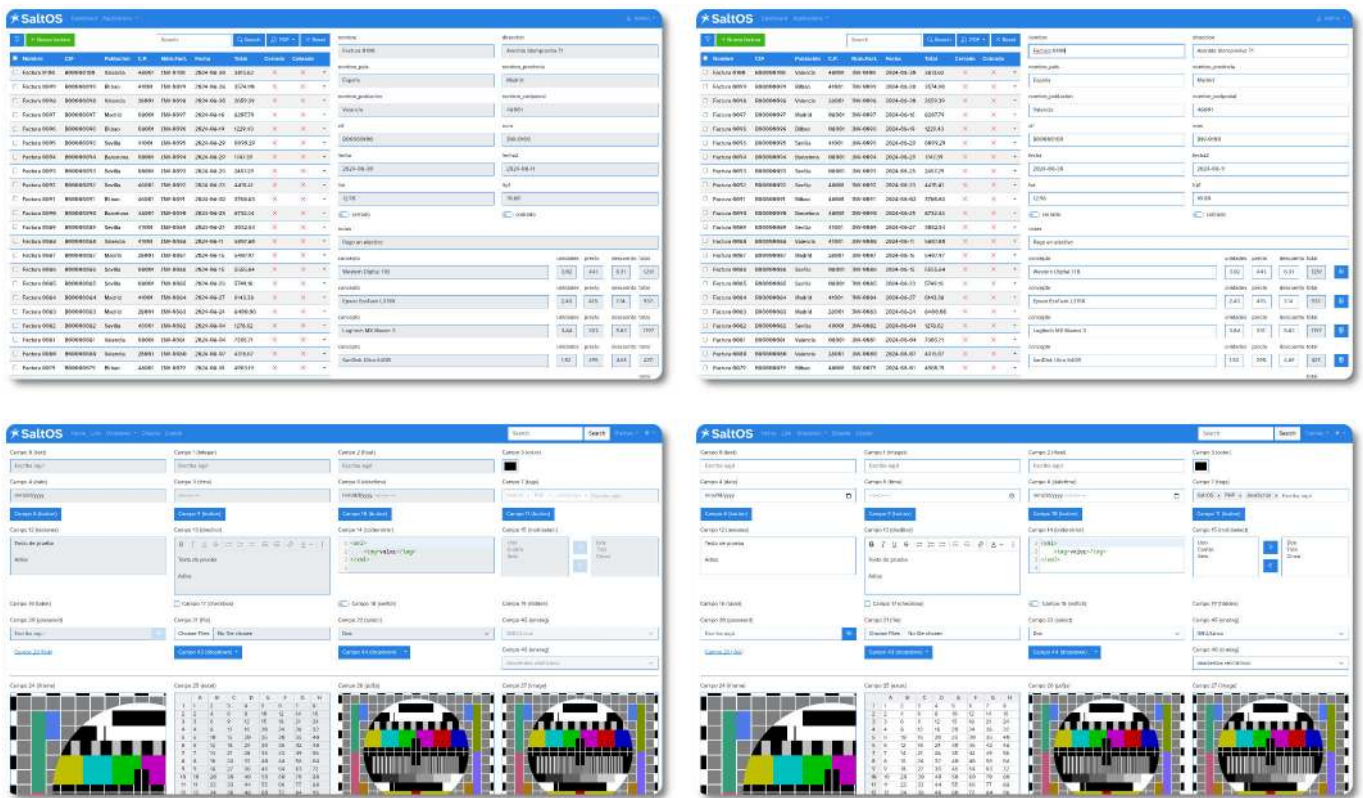
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7.4. AB Custom Transports & Logistics, S.L. (2019-2022)

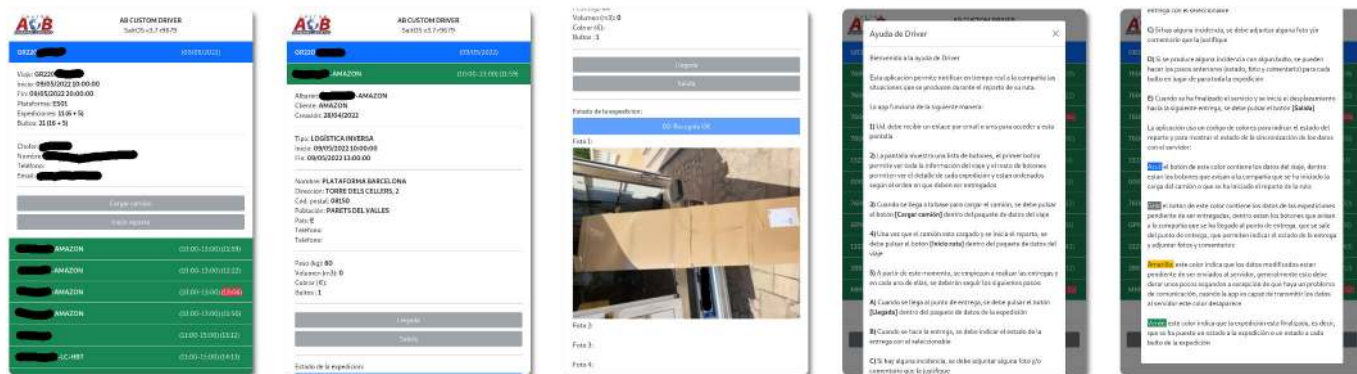
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- Automation of processes using embedded systems based on Raspberry Pi.

7.4.1. Last mile mobile application project

Develop an application that works on mobile phones so that the driver can do the last mile service:

- The application works with internet connection only for data loading, then it can work without internet
- The sending of information to the base is done asynchronously when there is an internet connection.
- It does not require installation, it is a 100% functional offline webapp using cache and local storage control techniques.
- Sending the route to the driver's device using email and/or SMS.
- Control of the arrival of the driver to the base and the loading of the truck before starting the routes.
- Sending automatic notices to recipients before going to each service.
- Control of the arrival, departure, delivery status, sending photos, comments, all this by expedition and/or by parcel.



7.4.2. Online appointment project

Development of the suite of applications so that recipients can make online appointments:

- Definition of platforms, delivery areas, distribution trucks and customers.
- Programming of expeditions, routes and call center applications.
- Definition of extras such as holidays, types of services, transfers, generic configuration, ...
- Integration of the app for making appointments with the company's website.
- Program the send of communications through emails and SMSs to the recipients.
- Generation of labels, delivery notes and route sheets for warehouses and traffic departments.



7.4.3. Label printers project using Raspberry Pi

Development of a pack based on label printer + Raspberry Pi for remote use:

- Connection to the host company network via network cable and DHCP.
- Allows the client to connect via FTP and WEB to the RPI to put import files.
- Automatic sending of import files to the central computer system.
- Reception and printing of the labels of the imported packages in real time.
- Ideal for customers and warehouses that must use labels of the delivery company.



ID	DESCRIPCION	CANTIDAD	UNIDAD	PRECIO	TOTAL	FECHA	ESTADO
123456789	PAQUETE DE 10 UNIDADES	10	UNIDADES	1000	10000	2023-10-27	ENVIADO
123456790	PAQUETE DE 20 UNIDADES	20	UNIDADES	1000	20000	2023-10-27	ENVIADO
123456791	PAQUETE DE 30 UNIDADES	30	UNIDADES	1000	30000	2023-10-27	ENVIADO
123456792	PAQUETE DE 40 UNIDADES	40	UNIDADES	1000	40000	2023-10-27	ENVIADO
123456793	PAQUETE DE 50 UNIDADES	50	UNIDADES	1000	50000	2023-10-27	ENVIADO
123456794	PAQUETE DE 60 UNIDADES	60	UNIDADES	1000	60000	2023-10-27	ENVIADO
123456795	PAQUETE DE 70 UNIDADES	70	UNIDADES	1000	70000	2023-10-27	ENVIADO
123456796	PAQUETE DE 80 UNIDADES	80	UNIDADES	1000	80000	2023-10-27	ENVIADO
123456797	PAQUETE DE 90 UNIDADES	90	UNIDADES	1000	90000	2023-10-27	ENVIADO
123456798	PAQUETE DE 100 UNIDADES	100	UNIDADES	1000	100000	2023-10-27	ENVIADO

7.4.4. SQLServer replica project in MariaDB

They must have a SQLServer replica in MariaDB to facilitate queries from SaltOS:

- Develop a driver in SaltOS to access SQLServer.
- Program system to initialize the desired tables.
- Monitor and control of the tables to detect inserts, updates and deletes in the SQLServer.
- Apply updates to MariaDB periodically and validate the integrity of the data.

7.4.5. Project upload documentation

Development of a mechanism that allows uploading documentation to the company's old ERP:

- Uploading PDF documents and images in JPEG and TIFF format with multiple layers.
- Documents can be identified if the file name contains the delivery note or the trip.
- It is possible to detect the delivery note or trip by means of the barcode that all documents carry.
- Errors in case of detection failure are reported in an error folder.
- Generation of records to trace errors and application to check the status of each document.

7.4.6. Client integrations project

System that allows the integration of clients with the generic import of the old ERP.

- System based on templates for mapping input data with those of the old ERP.
- Improvements in SaltOS import support (CSV, Excel, XML, Bytes and EDI files).
- Programming a query app to monitor the status of each import.
- Automatic generation of reports and fluxes outs for the return of information to customers.

7.5. Institut d'Estudis Espacials de Catalunya (2011-2019)

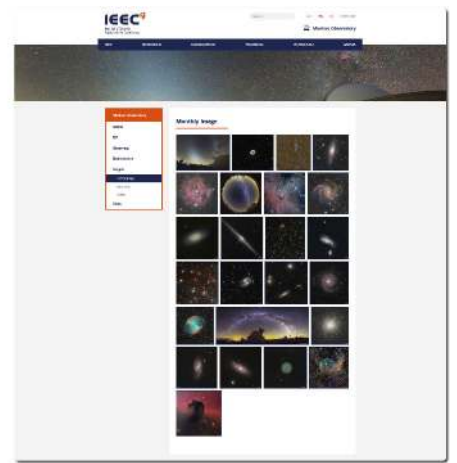
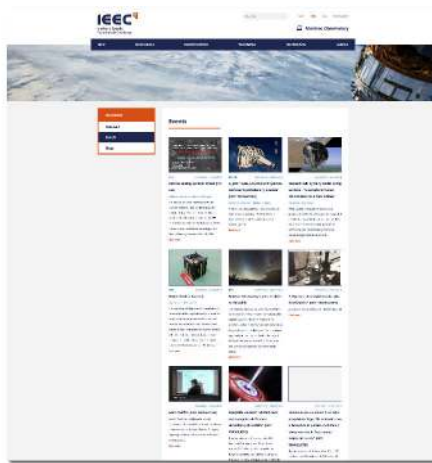
Developments for GNU/Linux environments to control robotic telescopes:

- Development of OpenROCS 2.0 (GPL-3.0 license), the control software used by the TJO and SQT telescopes.
- Development of the www.oadm.cat for the divulgation of the TJO telescope, the www.ice.csic.es for the ICE and the www.ieec.cat for the IEEC.
- Development of the MUR application: an online software for sending astronomical proposals.
- Fork of the IndiCCD project to allow the control of multiple Andor cameras.
- User interface and control system for the SQT telescope, the Allsky camera and the IEEC-CSIC telescope.

7.5.1. IEEC website project

Development of the IEEC website:

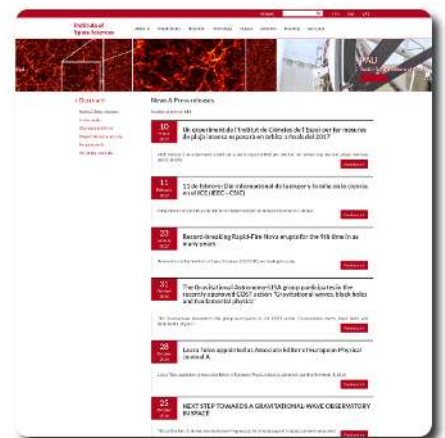
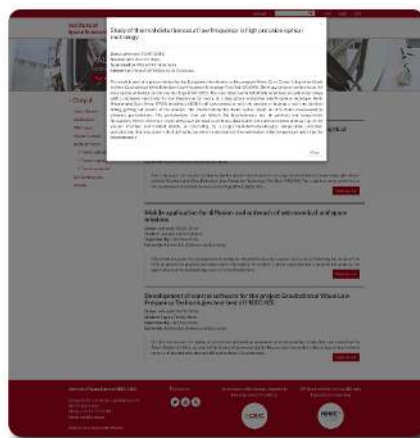
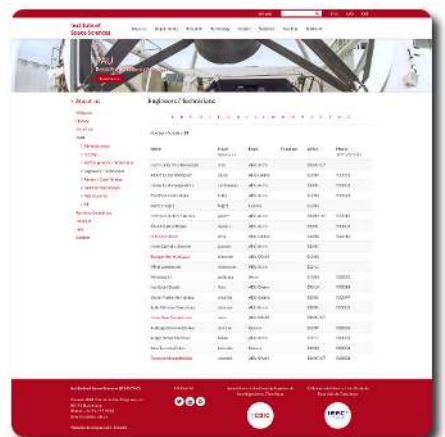
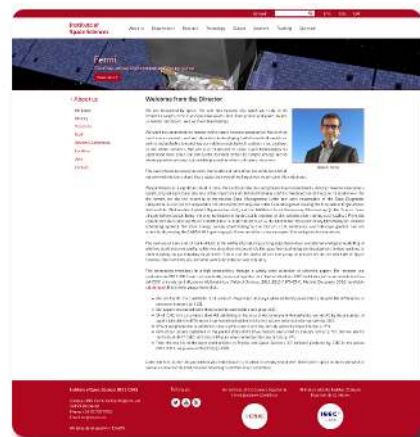
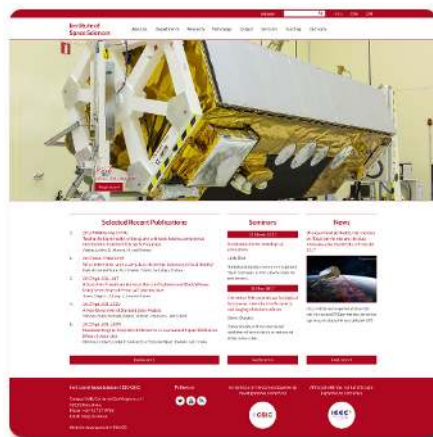
- Based on the RhinOS framework.
- Data synchronization using the databases of the units related to the institute.
- Responsive design that adapts to computers and mobile devices.
- WordPress content migration to the new website.
- <http://www.ieec.cat/>



7.5.2. ICE website project

Development of the ICE website:

- Based on the RhinOS framework.
- Data synchronization using the databases of the units related to the institute.
- Responsive design that adapts to computers and mobile devices.
- WordPress content migration to the new website.
- <http://www.ice.csic.es/>



7.5.3. IEEC-CSIC telescope project

Project to automate the IEEC-CSIC astronomical telescope:

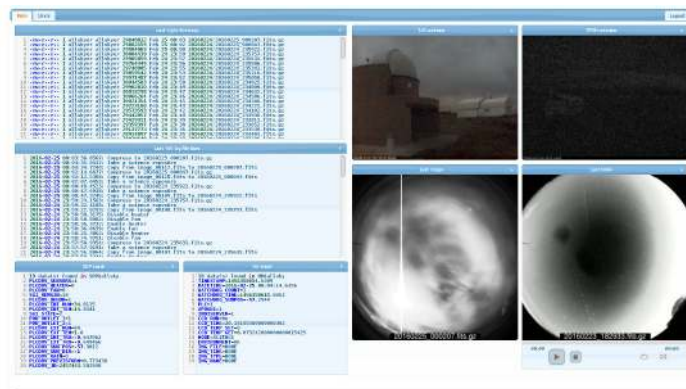
- Control system based in the OpenROCS 2.0 control software.
- Controlled devices using the industrial ROS standard.
- Development of the graphical user interface for control the entire system.
- Controlled devices: A Meade telescope, a Baader dome, an APC pdu, a Vaisala weather station, a cloud sensor and a SBIG camera.
- All system is controlled using only one Raspberry Pi.



7.5.4. Allsky camera project

Project to automate the IEEC Allsky camera:

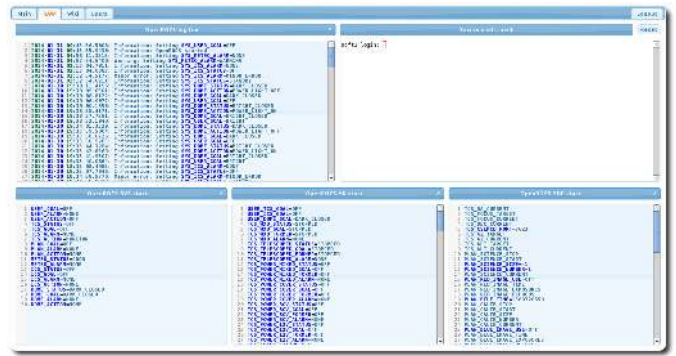
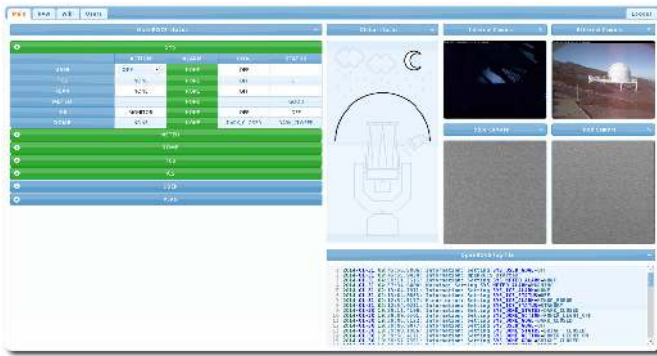
- Control system based in the OpenROCS 2.0 control software.
- Controlled devices using SNMP and INDI protocols.
- Programming of a driver that communicate with an industrial Siemens PLC.
- Development of the graphical user interface to monitor the entire system.
- Controlled devices: a heater, a fan, an APC pdu and an APOGEE camera.



7.5.5. SQT telescope project

Project to automate the SQT astronomical telescope:

- Installation of the operating systems in the two control computers (nominal and redundant).
- Development of some control packages as the dome control and the second motor focus.
- Installation of some specific software packages as talon, OpenROCS and IndiCCD to operate the telescope and instruments.
- Develop of the GUI to allow the remote supervision (useful in the commissioning process).



7.5.6. IndiCCD driver project

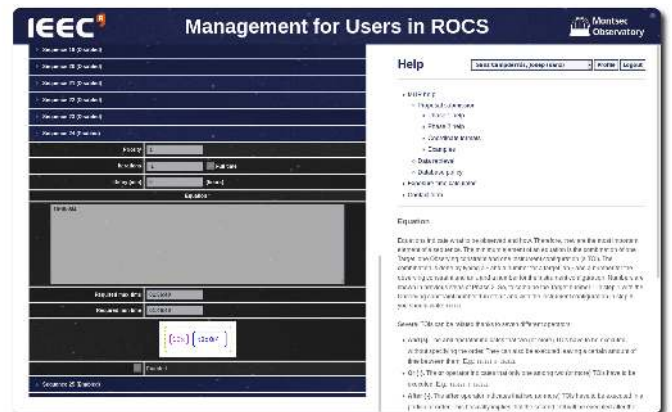
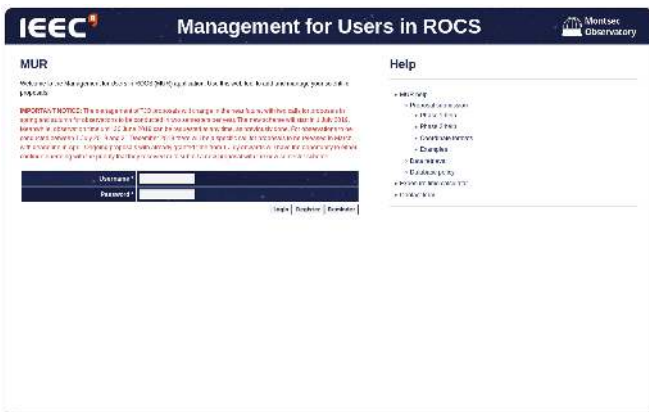
Fork of the XmCCD v.4.2.1 project:

- Added improvements to the cameras control (indiccd).
- Added support to the Andor and Finger Lakes Instruments cameras.
- Added support to use simultaneous cameras.

7.5.7. MUR project

Project that must allow the end users to send astronomical observing proposals:

- Design and implementation of the corporate website with private areas and user roles.
- Create a new broadcasting model for the OAdM webcams (using VLC) to prevent DoS.
- Data model specification to store the different phase data of the proposals.
- Language definition to allow end users to specify the observing sequences (using the toi concept).
- Implementation of tools to validate and plot the sequence equations.
- Front-end for each end user role that are involved in the different acceptance phases of the proposals (admin/CAT)



7.5.8. Projecto OAdM/TJO

Improvements in the infrastructure, and in the different software parts that are involved in the normal operation:

- Audit and automate execution of the PbCdIComm software for the data collection of the SMC weather station.
- Programming of drivers to communicate with the follow hardware equipments:
 - DAVIS Weather station.
 - Previstorm.

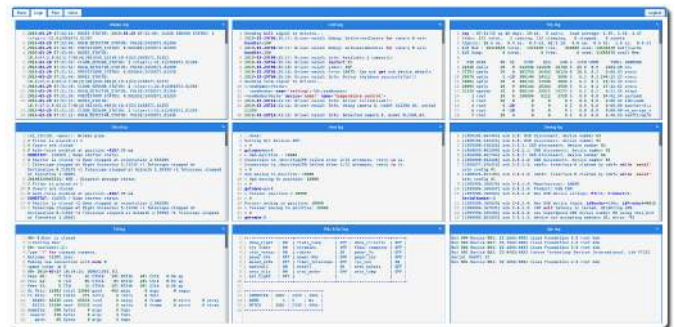
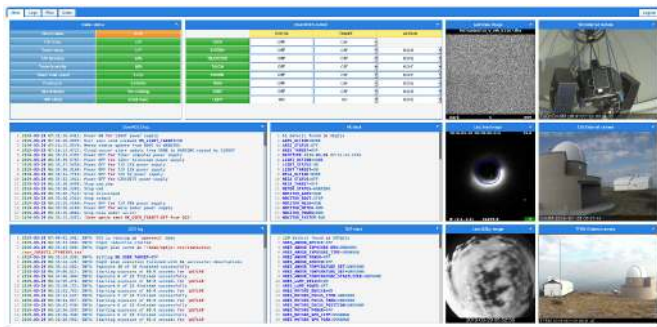
- Boltwood Cloud Sensor II (using the BWCloudSensorII code).
- Rain detector.
- Vaisala (using the Indi protocol).
- Configuration of the SNMP server to publish all data collected by the previous drivers.
- Improvements in the configuration and distribution of the network devices.
- <http://www.oadm.cat/>



7.5.9. OpenROCS v2.0 project

Software for unattended control of TJO (OAdM) and SQT astronomical telescopes.

- Design, implement and test control software using the following architecture.
- A server service that serves the requests from port 2323 to provide remote storage and accessible using predefined commands.
- A broadcast service that allows you to synchronize others remotely OpenROCS using broadcast techniques.
- A monitor service that the main task of this service is update the SDP and HK to be used by the scheduler.
- A scheduler service that lets you execute the actions that controls the telescope when you change the SDP or HK.
- This project is released under GPL-3.0
- <https://sourceforge.net/projects/openrocs/>
- <https://upcommons.upc.edu/handle/2099.1/26215>
- <https://github.com/josepsanzcamp/OpenROCS>



7.6. Wide Spectrum Software Solutions (2007-2011 in R+D Dept.)

Developments for GNU/Linux, MONO, Firmware, PDAs and web environments:

- Development of the SaltOS and RhinOS projects released under the GPL-3.0 license.
- Project for the NTE/ESA to do some parts of the unit-test of the LISA Pathfinder project.
- Project for the IEEC to make improvements in the GOLD-RTR project (v2 and v3).
- Multiple web projects (portals and online shops) to different customers.

7.6.1. NTE/ESA - LTPS project

SW HR for LTPS project:

- Unit-test tasks using ANSI C
- Validation-test tasks using Python

7.6.2. IEEC - GOLD_RTR v2.0 project

Improved control software for GOLD-RTR:

- Creation of 2 working roles (user and administrator)
- Modification of the predictive calculation model of GPS satellites
- Generation of the documentation

7.6.3. IEEC - GOLD_RTR live CD project

LiveCD with installer for easy tasks in campaigns:

- Allows working on LiveCD or installed with wizard on HDD.

- Useful for quick restore of the entire system.

7.6.4. IEEC - GOLD_RTR v3.0 project

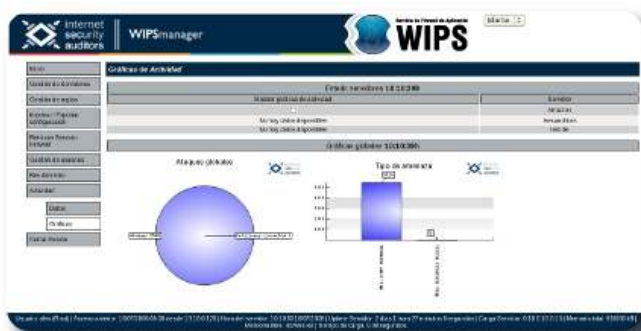
To carry out campaigns of experiments in Antarctica, it is necessary that the system is autonomous and can work remotely with and without communication in unattended mode:

- Improved control software GOLD-RTR
- Add a new algorithm for the integration of waveforms
- Validation of the new algorithm using unit-testing and validation-test.
- Generation of the documentation
- Deployment of a scheduler control system
- Management of the scheduler using pseudo-language files
- Integration with SVN commands to synchronize the results
- Notification system for monitoring the status of experiments
- Installation of the control system based on scheduler
- Generation of the documentation

7.6.5. ISEC Auditors - MANAGER WIPS project

Developing a security manager application for online servers:

- Import and management of the Apache logs server
- Generation of reports and security alarms
- Application that be used with multiple users, multiple servers and multiple roles.



The screenshot shows a detailed view of security events in the WIPSmanger interface. It includes a table with columns for 'Fecha de inicio', 'Fecha de fin', 'Tipo de evento', 'Usuario', 'Rol', and 'Estado'. The table lists several events with their respective dates and details. The interface is titled 'Internet security auditors' and 'WIPSmanger'.

7.6.6. AXA Winterthur - Tracking project

Several online applications:

- Follower for online campaigns (LAMP environment)
- Online shop for AXA Club

7.6.7. VUELING - PUNTO 2 project

Functional improvements project.

7.6.8. AIDA CENTRE - PDA HORMIPRESA project

Application for mobile devices that be used in process management:

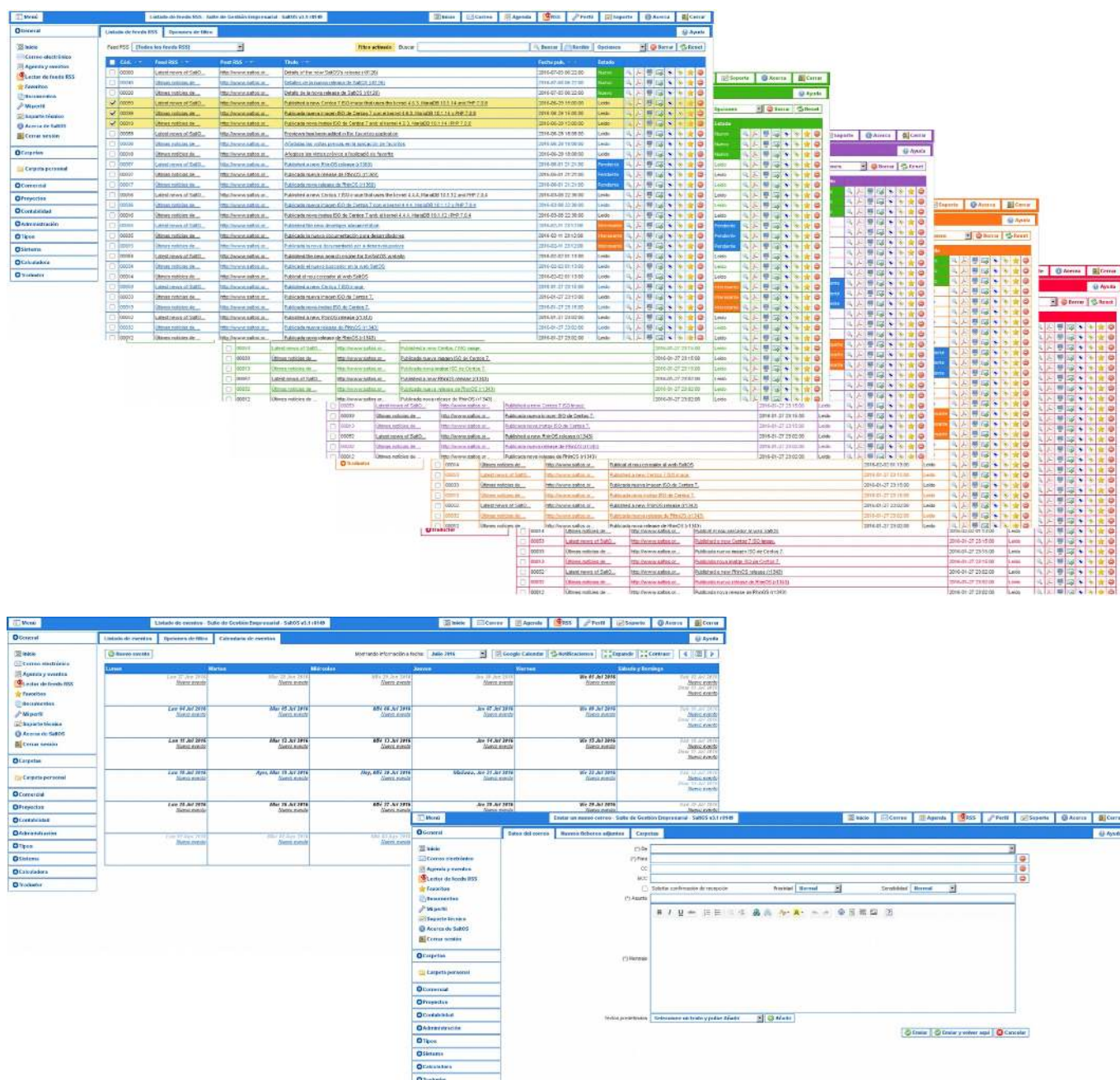
- Integration of RFID reader to read passive tags.
- Operating Modes using GPRS and WIFI networks or desktop connectivity.
- Desktop application for data synchronization

The image displays four screenshots of a PDA application interface for process management. The first screenshot shows a main menu with options: 'Notificación de Ferralla', 'Producción de Piezas', 'Control QA Ferralla', and 'Control QA Piezas'. The second screenshot shows the 'Notificación Ferralla' screen with fields for 'Fecha' (16/07/08), 'Colla' (01234567), 'Lote' (2), and 'Material' (22222222222222222222), along with 'Eliminar', 'Modificar', and 'Añadir' buttons. The third screenshot shows the 'Notificación Pieza' screen with fields for 'Fecha' (16/07/08), 'Sección' (SECCIO1), 'Molde' (MOTLLO01), 'Lote' (2), and 'Material' (22222222222222222222), also with 'Eliminar', 'Modificar', and 'Añadir' buttons. The fourth screenshot shows a detailed production screen with fields for 'Fecha' (16/07/08), 'Lote' (2), 'Material' (22222222222222222222), 'Est. antes' (BLOC), 'Est. desp.' (BLOC), 'Causa' (Desconocido), 'Acción' (Acción 1), and 'Texto'. It includes 'Eliminar', 'Modificar', and 'Añadir' buttons, and a table with columns: 'Fecha', 'Material', 'Est.Ini.', 'Est.Fin.', and 'Env.'.

7.6.9. WS3 - SaltOS Project (www.saltos.org)

Development of a framework for developing Rich Internet Applications:

- Based on XML + XSL technology on LAMP environments.
- Integration of management system for SMEs.
- Management of permissions on each system layer.
- Applications as: mail client, documentary manager, RSS and ATOM client, and more...
- User interface based in jQuery UI
- This project is released under GPL-3.0
- I continue developing this project on my own.
- <https://sourceforge.net/projects/saltos/>
- <https://github.com/josepsanzcamp/SaltOS>



7.6.10. WS3 - RhinOS Project (www.saltos.org/rhinos)

Development of a framework for developing web sites:

- Designed for LAMP environments.
- Includes 2 layers (CMS and CPS).
- Pseudo-code for the interpretation of the RhinOS templates.
- Separation of logical layers (presentation and business)
- This project is released under GPL-3.0
- I continue developing this project on my own.
- <https://sourceforge.net/projects/rhinos/>
- <https://github.com/josepsanzcamp/RhinOS>



7.6.11. CVA - SISAC project

Project for the Catalonia Agency of Water

- Its objective is service to the citizen.
- Based on eZpublish.
- Pilot project for 2000 users.

7.6.12. IEEC - GOLD-PRO project

Software develop for embedded system

- Use of XILINX hardware
- Programming MicroBlaze processors
- Programming LEON3 processor (space-certification)
- Routing Ethernet communications at low level (raw data)

7.6.13. RETEVISION - TRACEBOX Audit

Audit the software developed by HYR, TraceBox, that has been customized to service a client of Abertis Telecom:

- Identify the origin of the problems in streaming video.
- Propose a viable solution according to the current development.
- Generate a detailed technical report of the problem and proposals.
- Audit the correct deployment of the proposed solution and accepted

7.6.14. Alliaria (IN2 GROUP) - FIR@KEY project

Access control project for the fairs at the Fira de Barcelona:

- Development in VB6.0 using SOAP (by client requirements)
- Integration of RFID reader for user identification
- Installation wizard without dependencies for Microsoft operating systems.

7.6.15. Shop online CADAICO

Improvements to the initial online shop

7.6.16. Web site for the EUSS school

Web site and intranet for this university:

- Technology has been used RhinOS.
- There have been several evolutions with successful results.
- Customizing Moodle e-learning application

7.6.17. Web site for the COACB company

Functional enhancements of its corporate portals

7.6.18. Web site for the MEDIATORS OF INSURANCE ASSOCIATION

Custom deployment of an e-learning online tool.

7.6.19. Web site for the VERTEX company

Development of the corporate portal and intranet

7.6.20. Web site for the APQ company

Development of the corporate portal and intranet

7.6.21. Web site and online shop for the SEVIBE company

Several online applications:

- Development of corporate website
- Some promotional portals
- Development of online recruitment system and backoffice
- Integration of Wordpress News management.



7.6.22. Web site for the TRITON restaurant

Development of the corporate portal and intranet

- Online Reservation System

7.6.23. Web site for the Castelldetels's hostelry school

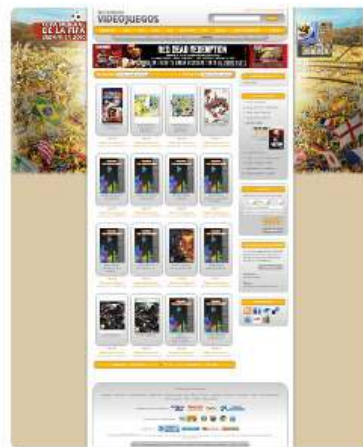
Development of the corporate portal and intranet



7.6.24. Online shop for the TUTIENDEVIDEOJUEGOS.COM company

Several online applications:

- Development of the corporate website, online shop and intranet
- System for mass data import
- System for mass mailings sending
- Management of personalized dispatching



7.7. Enfasystem (2006-2007 in R+D Dept.)

Developments for WEB using LAMP environments (Linux+Apache+MySQL+PHP):

- Improvement of the 'admin' control panel
- Development of the 'mecano' project
- Development of the DBMailer project
- Several collaborations with IIEC for the PARIS and GOLD-RTR projects

7.7.1. ADMIN project (www.saltos.org/rhinos)

Dynamic content manager for online applications

- Use of GPL software as TinyMCE and other

7.7.2. Mecano WEB project (www.saltos.org/rhinos)

System for generating dynamic web sites with the following characteristics:

- Programming using multi-tier architecture (templates and code)
- Design and implementation of high-level language
- Generation of thumbs in realtime (images)
- Generation of multimedia content in realtime (audio)
- Documentation generation in real time (PDF)

7.7.3. DBMailer project (www.saltos.org/rhinos)

System to sending mass mailings

- Controled by database
- Control panel for maintenance
- Trigger and checks via crontab for detection of errors and system crashes

7.7.4. EXIT Foundation - Social participation project

Portal of the social participation:

- Analysis and design of the database for the project
- Implementation of the control panel to administer and coordinate the fundation.
- Compliance with the Data Protection Act (Spanish Organic Law on Data Protection)
- Quality control for the processes of information

7.7.5. IEEC - GOLD_RTR project

Software for IEEC in the GOLD_RTR project:

- Segmentation of the configuration files
- Reorganization of the configuration lines according to the GPS data
- Improvements in the graphical interface

7.7.6. IEEC - PARIS signal processor subsystem project

Programming a NIOS2 microprocessor

- Design and implementation of a program to be used inside a ALTERA NIOS2 microprocessor
- Transmission of more that 4Mbytes/seg using an ethernet connection (UDP)
- Parser for evaluate commands and validation of these
- Optimization of the ALTERA libraries to improve timings
- Control via digital oscilloscope to monitor timings and guaranty the correct work.

Programming a software (for GNU/Linux platforms) for reception of UDP frames

- Design and implementation of a program for reception of UDP frames
- Validation of the frames and checking their integrity

- Saving data to local disk at high speed (using DMA technology)

Configuration of the computer systems for data reception

- Installation of GNU/Linux Slackware 11.0
- System Setup (remove unneeded processes)
- Software Installation for UDP frame reception
- Empirical calibration and performance certification document

7.8. Ingeniería de la Información, IN2 (2005-2006 as team's chief) ---

Software development for GNU/Linux and Windows CE.

- Use of security tools like Snort.
- Programming of mobile devices using EVC++ 3.0 for PDA devices
- Programming in C, PHP, .NET
- Use of DBMS such as MySQL, ORACLE

7.8.1. SNIFFER project ---

Capture and system control for network communications TETRA

- Using Snort software to capture IP frames
- Shell scripts using Python
- Download real-time data using FTP and HTTP protocols
- Generation of reports by using web technology.

7.8.2. IVP project ---

Urban maintenance software for PDA devices

- Using EVC++ 3.0 using GIS software Adobe OnSiteView
- Communication link using Bluetooth to GPS receivers and NMEA 0183 protocols for parse outputs.
- Using HP libraries to integrate camera inside the software
- Use of XML for IO transfers

7.8.3. WEB CONTROL project ---

Software for monitoring network status

- Using SNMP v1.0 and v2.0
- Programming in C for the daemon service
- PHP programming language for the user interface
- Using MySQL 4.0.23 for massive data storage (daemon and web)

7.8.4. SICAT project ---

Software demonstration of TETRA network operation

- Wizard for DB module design
- DB module WebService
- Using MySQL for DB module
- Terminal Module (TETRA terminal emulation)
- Module DB (dynamic application defined by Wizard)
- GIS Module (programming using SDK's for TomTom GPS network)

7.8.5. Upgrades for some applications at BAMESA

Development of applications software to migrate from Portal to PHP

- Table maintenance programming
- Generation of reports of various types: lists, graphs, calendars
- Conversion from DB triggers to PHP code
- Using AJAX to create dynamic filters
- Import and export of all filters
- Using the ORACLE DBMS

7.9. Institut d'Estudis Espacials de Catalunya (2001-2005)

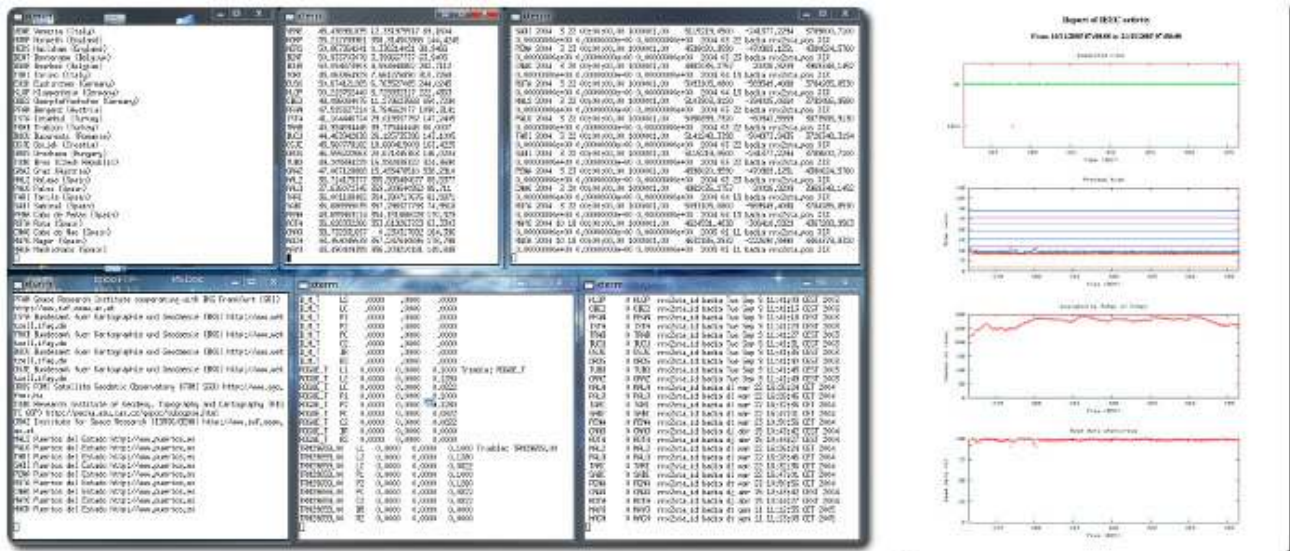
Development of software for UNIX environments.

- Design and implementation of hardware and drivers.
- Planning and implementation of software for various projects.
- Programming in C with GTK, Tcl/Tk on Unix tcsh.

7.9.1. REALTIME (COST716) project

Automation of the calculation of Total Zenith Delay

- Shell scripts using tcsh, tcl/tk, etc ...
- Download realtime data using FTP, HTTP, SSH
- Generation of 'reports' by using web technology and LaTeX.
- Management processes on the Linux kernel.
- Detection of bugs in the file system Solaris 2.6



GUI for RealTime by SANZ - Rinex		Real stations for DDG MS	
File	View About	Station	Year
FTP DIR (loc)	ftp://loc.ucsd.edu/pub/rinex/years/	loc	2004
FTP_RAW (loc)	ftp://loc.ucsd.edu/pub/rinex/years/	loc	2004
FTP_FILE (loc)	ftp://loc.ucsd.edu/pub/rinex/years/	loc	2004
FTP_DIR (icc)	ftp://ftp.icc.es/gps/stations/gps_stations/rinex	icc	2004
FTP_RAW (icc)	ftp://ftp.icc.es/gps/stations/gps_stations/rinex	icc	2004
FTP_FILE (icc)	ftp://ftp.icc.es/gps/stations/gps_stations/rinex	icc	2004
FTP_DIR (icc2)	ftp://ftp.icc.es/gps/stations/gps_stations/rinex	icc2	2004
FTP_RAW (icc2)	ftp://ftp.icc.es/gps/stations/gps_stations/rinex	icc2	2004
FTP_FILE (icc2)	ftp://ftp.icc.es/gps/stations/gps_stations/rinex	icc2	2004
FTP_DIR (igs)	ftp://igs.ensg.ign.fr/pub/gps/rinex/years/	igs	2004
FTP_RAW (igs)	ftp://igs.ensg.ign.fr/pub/gps/rinex/years/	igs	2004
FTP_FILE (igs)	ftp://igs.ensg.ign.fr/pub/gps/rinex/years/	igs	2004
FTP_DIR (gpc)	ftp://ftp.ii.ac.es/out/gpc/gpc	gpc	2004
FTP_RAW (gpc)	ftp://ftp.ii.ac.es/out/gpc/gpc	gpc	2004
FTP_FILE (gpc)	ftp://ftp.ii.ac.es/out/gpc/gpc	gpc	2004
FTP_DIR (loc)	ftp://rinex.ii.ac.es/out/gpc/gpc	loc	2004
FTP_RAW (loc)	ftp://rinex.ii.ac.es/out/gpc/gpc	loc	2004
FTP_FILE (loc)	ftp://rinex.ii.ac.es/out/gpc/gpc	loc	2004

7.9.2. DD_RCVR and LOTTOS projects

Software for processing mass data using shell scripts

- Graphical interfaces for scientific use

7.9.3. GRACE-II project

Calibration software for the altimeter of ENVISAT satellite

- Graphical interface for use by data processing operators
- System for automated data transfer to the ICM institute

7.9.4. STD/IEEC Group Intranet

Specification and implementation of the online portal:

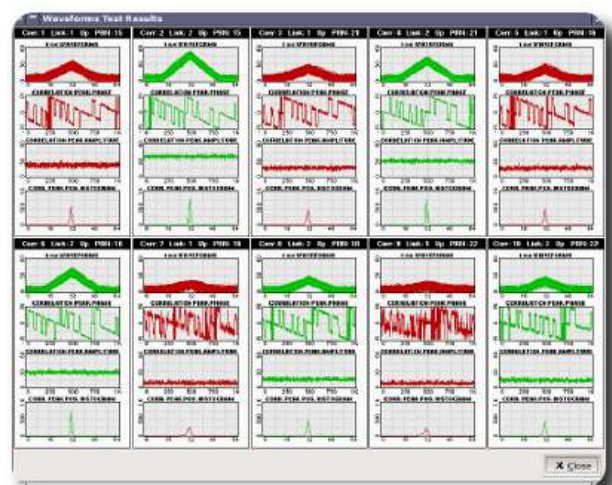
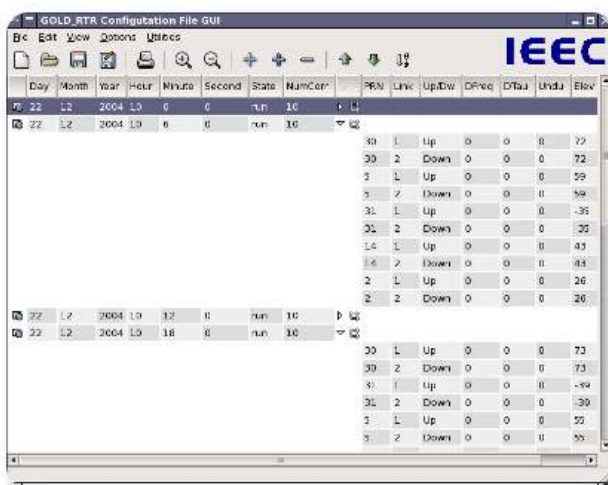
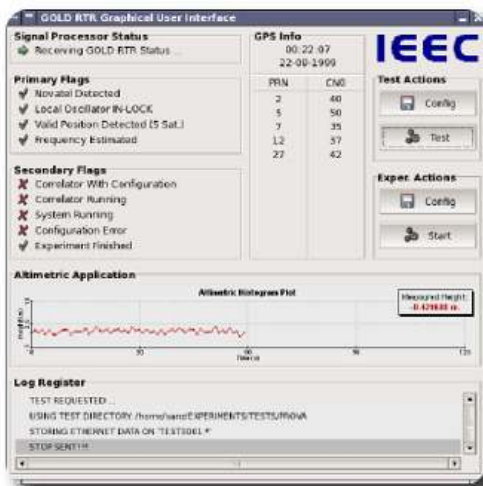
- Using by web technology such as PHP

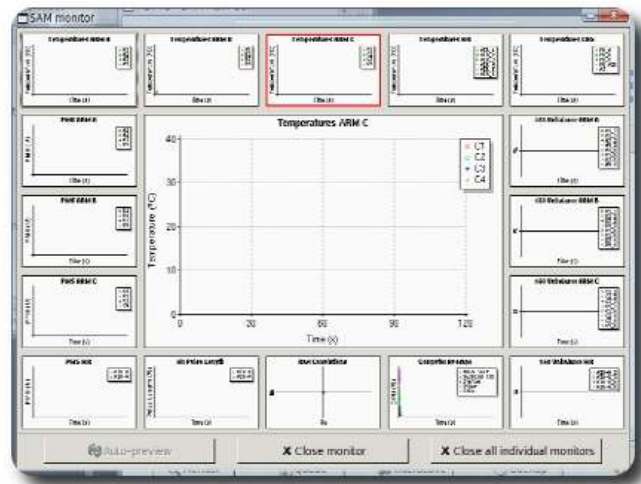
- User Authentication
- Use secure protocols (SSL) on apache
- Accessing Data with MySQL and PostgreSQL

7.9.5. GOLD_RTR (PARIS) and MDPP3 (SMOS) projects

for ALTERA programming environments:

- Communications with GPS receivers (Novatel, Ashtech).
- Software and hardware for ALTERA development boards
- Multiprocessor systems and realtime applications.
- Specification of ALTERA NIOS2 microprocessors
- Software for GNU/Linux in C and GTK
- Design of communication protocols over UDP/IP
- Transmission of data in realtime mode
- Reception and realtime storage data (12 Mbytes/sec)
- System monitoring graph in realtime mode





7.10. Univ. Politècnica de Catalunya, Dept. of Applied Physics (1999-2001)

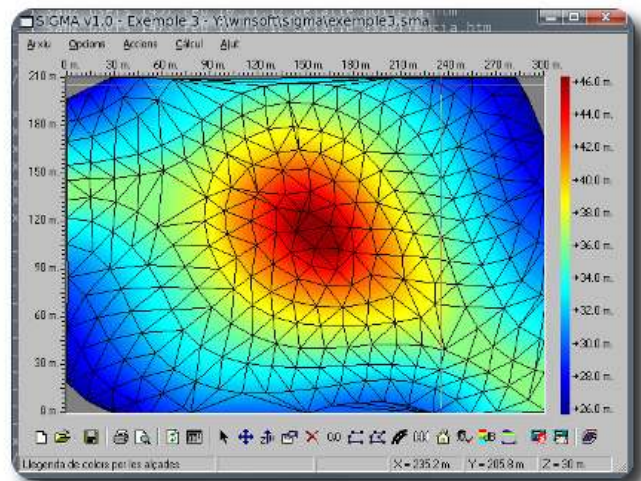
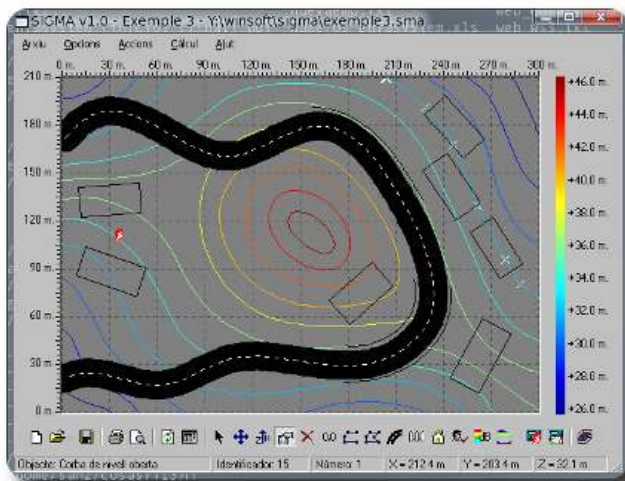
Administration of Unix Systems and Microsoft Windows

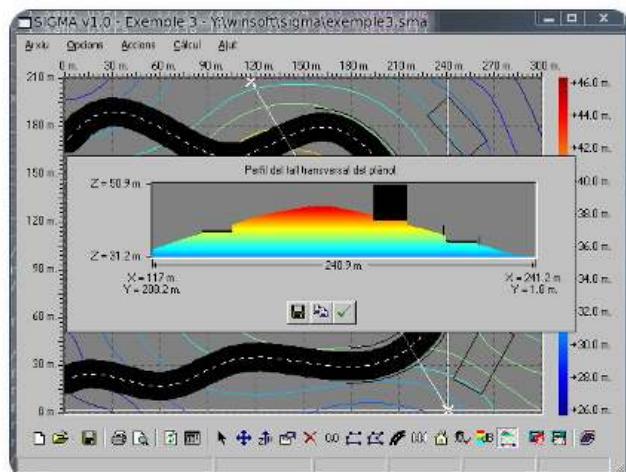
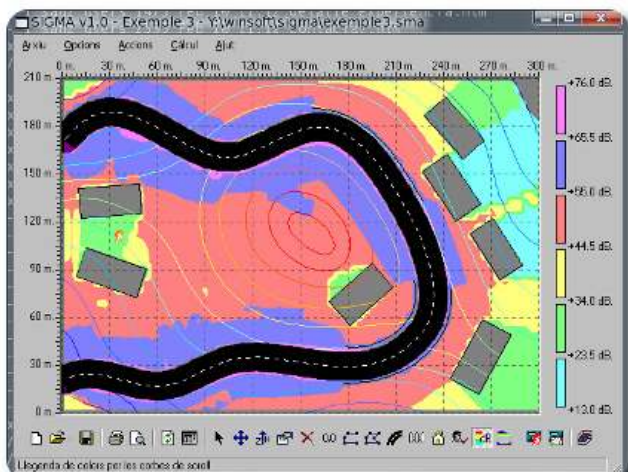
- Implementation of network security systems (SSL)
- Security Audits of computer systems department
- Software development and maintenance of the department's intranet
- Software development for the Generalitat de Catalunya (SIGMA and NivAval)
- Collaborations with the WFAE association.
- Support to users, and so on.

7.10.1. SIGMA, Sistema Informàtic de Gestio i Modelització Acústica

Project of an application with graphical environment for calculating levels of noise pollution:

- Develop an easy and intuitive graphical interface, using Visual Basic, to provide to the user all the necessary elements to enter the input data to the model.
- Develop a set of libraries programmed in Visual C++ to provide all the computing power and speed that only C can offer.
- To perform the topographical simulations and obtain a valid model for the model calculation, a 3D representation mechanisms were implemented using Delaunay triangulation and cubic spline interpolation.





7.10.2. NivAval, Nivell d'Avaluació Lar

Project to evaluate the results of acquisition campaigns of noise levels using sound level meters:

- Develop a graphical interface using Visual Basic for data entry.
- Implement the model defined in the law 16/2002 of protection of noise pollution.
- Generating reports in text and graphical format highlighting the relevant information established in each case by the acoustic law.

